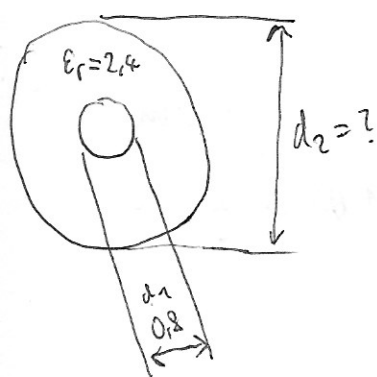


Koaxiální kabel má průměr vnitřního vodiče $d_1 = 0,8 \text{ mm}$ a dielektrikum zhotovené z polyethylénu s $\epsilon_r = 2,4$. Pro pracovní kmitočet, pro který je kabel určen, lze považovat za ideální bezztrátové vedení

- a) Jak velký musí být poměr vnějšího vodiče, aby byla charakteristická impedance vedení $Z_0 = 50 \Omega$.
- b) Úsek výše uvedeného coax. vedení o délce $x = 0,2 \text{ m}$ je na konci spojený na krátko. Pro pracovní kmitočet se z hlediska vstupních svorek bude jevit jako nekonečná impedance. O jaký kmitočet se jedná?
- c) Jak velká při kmitočtu z bodu b) skupinová rychlost vlny vedené po tomto vedení?
- d) Jak by se z hlediska vstupních svorek impedance jevil stejně dlouhý úsek vedení kdyby byl na konci rozpojený?



$$\text{a) } Z_0 = \frac{60}{\sqrt{\epsilon_r}} \ln\left(\frac{d_2}{d_1}\right)$$

$$50 = \frac{60}{\sqrt{2,4}} \ln\left(\frac{d_2}{0,8}\right)$$

$$0,8 e^{\frac{50}{60} \sqrt{2,4}} = d_2 = 2,909 \text{ mm}$$

$$\text{b) } Z_{VST} = Z_0 \frac{Z_k + j Z_0 \tan(\beta l)}{Z_0 + j Z_k \tan(\beta l)}$$

$$Z_{VST} \stackrel{Z_k=0}{=} j Z_0 \tan\left(2\pi \frac{l}{\lambda}\right)$$

$$\text{Pro } l = \frac{\lambda}{4} \dots Z_{VST} \rightarrow \infty$$

$$l = 0,2 \text{ m} = \frac{\lambda}{4}$$

$$\lambda = 0,8 \text{ m}$$

$$f = \frac{c}{\lambda \sqrt{\epsilon_r}} = \frac{3 \cdot 10^8}{0,8 \cdot \sqrt{2,4}} = 242 \text{ MHz}$$

$$f_{krit} = \frac{c}{\pi \sqrt{\epsilon_r}} \left(\frac{1}{1 + \frac{\lambda}{2}} \right)$$

$$= \frac{3 \cdot 10^8}{\pi \sqrt{2,4}} \left(\frac{1}{1 + \frac{0,8}{0,8}} \right) = 33,3 \text{ MHz}$$

$$\text{c) } f = 242 \text{ MHz}$$

$$V_g = \frac{c}{N_g} =$$

$$N_g = N_F = \frac{c}{\sqrt{\epsilon_r}} = \frac{3 \cdot 10^8}{\sqrt{2,4}} = 1,93 \cdot 10^8 \text{ m s}^{-1}$$

$$N_g \approx \frac{c}{\sqrt{\epsilon_r}} \approx 1,93 \cdot 10^8 \text{ m s}^{-1}$$

$$\text{d) } Z_{VST} = Z_0 \frac{Z_k + j Z_0 \tan(\beta l)}{Z_0 + j Z_k \tan(\beta l)} \stackrel{L'H}{=} Z_0 \frac{1}{j \tan(\beta l)} = 0$$

Vední o délce 0,5 m je tvořeno koaxiálním kabelem s průměrem vnitřního vodiče $a = 1 \text{ mm}$ a většího vodiče 6 mm. Koaxiální kabel má dielektrikum s relativní $\epsilon_r = 2,1$.
Nakonec rozpojme, dochází k total odrazu.

a) $Z_{VST} = 0$ $F = ?$

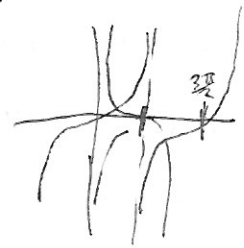
b) Kolik procent výkonu by prošlo do zátěže $R = 10 \Omega$

c) Jakou impedanci musí mít vední, aby se nic neodráželo?

$a = 1 \text{ mm}$
 $b = 6 \text{ mm}$

$$Z_0 = \frac{60}{\sqrt{\epsilon_r}} \ln\left(\frac{b}{a}\right) = \frac{60}{\sqrt{2,1}} \ln\left(\frac{6}{1}\right) = 74,2 \Omega$$

$$Z_{VST} = -j Z_0 \cot\left(\frac{2\pi l}{\lambda}\right) = 0$$



$$\frac{2\pi l}{\lambda} = \arctan(1)$$

$$\frac{2\pi l}{\lambda} = \frac{\pi}{2} \Rightarrow l = \frac{\lambda}{4} = \frac{0,5}{4} = 0,125 \text{ m}$$

$$f = \frac{c}{\lambda} = \frac{3 \cdot 10^8}{0,25} = 1,2 \text{ GHz}$$

a každé liché násobky

~~$R = 10 - j28,7$~~

~~$R = 10 + j28,7$~~

~~$R = \frac{10 - j28,7}{10 + j28,7}$~~

~~$T = 1 + R = 0,52$~~

~~$\% P_{proch} = 1 - |R|^2 = 0,42 = 42\%$~~

~~$\frac{P_{in}}{P_{out}} = \frac{1}{|T|^2} \frac{Z_K}{Z_0} = 0,52^2 \frac{10}{28,7} =$~~

~~$\frac{P_{out}}{P_{in}} = \frac{|R|^2}{28,7^2} = \frac{10^2}{28,7^2}$~~

c) $Z_0 = Z_K = 74,2 \Omega$ nic se nebude odrážet

KOAX 3

Koaxiální kabel má vnitřní vodič o průměru 1 mm $\epsilon_r = 2,2$

a) $b = ?$ $Z_0 = 75 \Omega$

$$Z_0 = \frac{60}{\sqrt{\epsilon_r}} \ln\left(\frac{b}{a}\right) = \frac{60}{\sqrt{2,2}} \ln\left(\frac{b}{1}\right) = 75$$

b) $f = 400 \text{ MHz}$ $Z_0 = -j75 \Omega$ $l = ?$

a) $b = e^{\frac{75\sqrt{2,2}}{60}} = 6,4 \text{ mm}$ ✓

c) E_m v koaxu $S_{\text{STR}} = 5 \text{ W}$

b) $\lambda = \frac{c}{f\sqrt{\epsilon_r}} = \frac{3 \cdot 10^8}{400 \cdot 10^6 \sqrt{2,2}} = 0,506 \text{ m}$

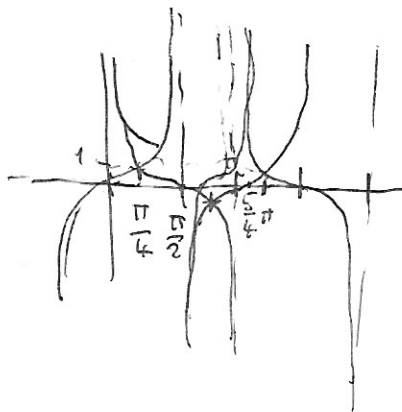
S otevřeným koncem $Z_k \rightarrow \infty$

$$Z_{\text{vst}} = -jZ_0 \cot\left(\frac{2\pi l}{\lambda}\right)$$

$$\cot\left(\frac{2\pi l}{\lambda}\right) = 1$$

$$\frac{\pi}{4} = \frac{2\pi}{\lambda} l$$

$$\frac{\lambda}{8} = l = 6,3 \text{ cm}$$



Se zkrat koncem $Z_k = 0$

$$Z_{\text{vst}} = Z_0 \frac{jZ_0 \tan\left(\frac{2\pi l}{\lambda}\right)}{Z_0 + jZ_0 \tan\left(\frac{2\pi l}{\lambda}\right)} = jZ_0 \tan\left(\frac{2\pi l}{\lambda}\right)$$

$$\rightarrow l = \frac{\lambda}{8} + \frac{\lambda}{2} \cdot k \quad (k = 0, 1, 2, 3, \dots)$$

$$= 6,3 \text{ cm} + (25 \text{ cm} \cdot k)$$

c) $P_m = \frac{1}{2} \frac{|U_m|^2}{|Z|} \Rightarrow U_m = \sqrt{2S_m Z_0}$

$$U_m = \sqrt{2 \cdot 5 \cdot 75} = 27,4 \text{ V}$$

$$E_m = \frac{U}{a \ln\left(\frac{b}{a}\right)} = \frac{27,4}{\ln(6,4)} = 2052 \text{ V/m}$$

$$\frac{27,4}{10 \ln(6,4)} = 14,76 \text{ V/m}$$

Vlnovody rezonátory



Nauknete obdeltkový vlnovod $a, b; a > b$; pracovní $F = 3 \text{ GHz}$

= střed pásma s dominantním módem
a druhé dva nejbližší vyšší módy,

kteří připodají vlivu, byly stejně tlumeny
Vlnovod je vyplněn suchým vzduchem.

a) Rozměry? $a = ?$ $b = ?$

b) $\lambda = ?$

c) Tlumení pro dva nejbližší módy?

$$a = 0,025 \text{ m}$$

$$b = 0,0125 \text{ m}$$

Pracovní pásmo

třeba 1-5 GHz

$$\frac{3}{4} \frac{c}{a} = F$$

$$a = \frac{3}{4} \frac{c}{F} = \frac{3}{4} \frac{3 \cdot 10^8}{3 \cdot 10^9} = 0,075 \text{ m}$$

$$b = \frac{a}{2} = 0,0375 \text{ m}$$

TE₁₀ - dominantní mód

$$f_{\text{dom}} = \frac{c}{2a} \quad f_{\text{rez}} = \frac{c}{a} \quad f_s = \frac{f_m + f_d}{2} = \frac{\frac{c}{2a} + \frac{c}{a}}{2} = \frac{3c}{4a}$$

$$f_{\text{střed}} = 3 \text{ GHz}$$

$$f_{\text{krit}(2,0)} = \frac{c}{2\sqrt{\epsilon_r}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$f_{\text{krit}(2,0)} = 1,5 \cdot 10^8 \sqrt{\left(\frac{2}{0,025}\right)^2 + 0^2} = 1,2 \text{ GHz}$$

$$f_{\text{krit}(0,1)} = 1,5 \cdot 10^8 \sqrt{0^2 + \left(\frac{1}{0,0125}\right)^2} = 12 \cdot 10^9$$

$$b = \frac{1,5 \cdot 10^8}{12 \cdot 10^9} = 0,0125 \text{ m} = 12,5 \text{ mm}$$

$$k = \frac{\omega}{c\sqrt{\epsilon_r}} = \frac{2\pi \cdot 3 \cdot 10^9}{3 \cdot 10^8} = 62,83$$

$$\text{TE}_{0,1} \quad k_x = \frac{m\pi}{a} = 0 \quad k_y = \frac{n\pi}{b} = \frac{\pi}{0,0375} = 83,78$$

$$k_z = \sqrt{k^2 - k_x^2 - k_y^2} = \sqrt{62,83^2 - 0^2 - 83,78^2} = j55,42$$

$$\text{TE}_{2,0} \quad k_x = \frac{m\pi}{a} = \frac{2\pi}{0,075} = 83,78$$

$$k_y = 0$$

$$k_z = \sqrt{k^2 - k_x^2 - k_y^2} = \sqrt{62,83^2 - 83,78^2} = j55,42$$

Oba módy mají
útlum $e^{-55,42z}$ ✓

Obdélníkový kovový vlnovod o rozměrech $a=30\text{ mm}$ a $b=15\text{ mm}$ je buzen signálem o kmitočtu $7,5\text{ GHz}$

a) F_{krit} pro vidy řád $(m,n=2)$

b) Pásmo jednovidosti

c) Vede vlnovod při $F=7,5\text{ GHz}$ dále? Jak jsou tlumeny dva vyšší módy

$$f_{\text{krit}1,0} = \frac{c}{2\sqrt{\epsilon_r}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} = \frac{3 \cdot 10^8}{2} \sqrt{\left(\frac{1}{0,03}\right)^2 + 0} = \frac{3 \cdot 10^8}{0,03} = 10\text{ GHz}$$

$$f_{\text{krit}0,1} = \frac{c}{2\sqrt{\epsilon_r}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} = \frac{3 \cdot 10^8}{2} \sqrt{0 + \left(\frac{1}{0,015}\right)^2} = \frac{3 \cdot 10^8}{0,015} = 20\text{ GHz}$$

$$f_{\text{krit}1,1} = \frac{c}{2\sqrt{\epsilon_r}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} = \frac{3 \cdot 10^8}{2} \sqrt{\left(\frac{1}{0,03}\right)^2 + \left(\frac{1}{0,015}\right)^2} = 22,36\text{ GHz}$$

$$f_{\text{krit}2,0} = \frac{3 \cdot 10^8}{2} \sqrt{\left(\frac{2}{0,03}\right)^2} = 20\text{ GHz}$$

$$f_{\text{krit}2,1} = 22,36\text{ GHz}$$

$$f_{\text{krit}1,2} = 20,62\text{ GHz}$$

$$f_{\text{krit}2,2} = 22,36\text{ GHz}$$

b) Pásmo zjed $f_{\text{krit}1,0} \rightarrow f_{\text{krit}0,1}$ $= 10 - 20\text{ GHz}$ $k = \frac{\omega}{c} \sqrt{\epsilon_r} = \frac{2 \cdot \pi \cdot 7,5 \cdot 10^9}{3 \cdot 10^8} \cdot 1 = 157,1\text{ m}^{-1}$

c) vid $TE_{1,0}$ na $F=7,5\text{ GHz}$ dále

$$k_z(TE_{1,0}) = \sqrt{k^2 - k_x^2 - k_y^2} \quad k_{x_{1,0}} = \frac{m\pi}{a} = \frac{2\pi}{0,03} = 209,4\text{ m}^{-1} \quad k_{y_{1,0}} = 0 \quad k_z = j 138,4 \cdot 10^2$$

$$k_z(TE_{1,0}) = \sqrt{k^2 - k_x^2 - k_y^2} = j 138,4$$

$$k_x = 0$$

$$k_y = \frac{n\pi}{b} = \frac{1\pi}{0,015} = 209,4$$

$$TE_{1,0} \text{ a } TE_{0,1} \text{ jsou tlumeny } e^{-k_z z} = e^{-138,4 z}$$

Navrhněte vlnovod $\square$ vid TE_{10} $F = 7,5 \text{ GHz}$, F je střed pásma 1-vídní přenos.
Nejbližší vyšší módy stejné tlumeny

$$f = \frac{f_d + f_h}{2} = \frac{\frac{c}{2a} + \frac{c}{2b}}{2} = 7,5 \cdot 10^9$$

$$\therefore \frac{3c}{2a} = 15 \cdot 10^9$$

$$a = \frac{3}{4} \frac{c}{F} = \frac{3 \cdot 3 \cdot 10^8}{4 \cdot 7,5 \cdot 10^9} = 0,03 \text{ m}$$

$$f_d = \frac{c}{2a} = \frac{3 \cdot 10^8}{2 \cdot 0,03} = 5 \cdot 10^9 \text{ Hz } (TE_{1,0})$$

$$f_h = \frac{c}{b} = 10 \text{ GHz } (TE_{0,1})$$

$$b = \frac{a}{2} = 0,015 \text{ m}$$

Jak velký P lze přenášet $\square$ vid TE_{10} $F = 7,5 \text{ GHz}$ $E_{\max} = 3 \frac{\text{kV}}{\text{mm}}$
 $a = 30 \text{ mm}$, $b = 15 \text{ mm}$

$$E_{\max} = 3 \cdot 10^6 \frac{\text{V}}{\text{m}}$$

Příčné konst.

$$k_x = \frac{\pi}{a} = \frac{1\pi}{0,030} = 1,047 \cdot 10^2 = 104,7 \text{ m}^{-1}$$

$$k = \frac{\omega}{c} \sqrt{\epsilon_r} = \frac{2\pi F}{c} \sqrt{\epsilon_r} = \frac{2\pi \cdot 7,5 \cdot 10^9}{3 \cdot 10^8} = 157,1 \text{ m}^{-1}$$

$$k_z = \sqrt{k^2 - k_x^2 - k_y^2} = \sqrt{157,1^2 - 104,7^2} = 117,1$$

$$Z_{TE} = \frac{\omega \mu_0}{k_z} = \frac{2\pi \cdot 7,5 \cdot 10^9 \cdot 4\pi \cdot 10^{-7}}{117,1} = 505,7 \Omega$$

$$P = \frac{a \cdot b}{4} \cdot \frac{E_{\max}^2}{Z_{TE}} = \frac{0,03 \cdot 0,015}{4} \cdot \frac{(3 \cdot 10^6)^2}{505,7} = 15,02 \cdot 10^6 \text{ W} = 15,02 \text{ MW}$$

Obdélníkový rezonátor

$$a = 0,06 \text{ m} \quad b = 0,03 \text{ m}$$

$$f_{\text{REZ}} = 10 \cdot 10^9 \text{ Hz} = 10 \text{ GHz}$$

Mod TE₁₀₁

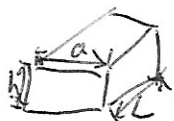
$$m=1 \quad n=0 \quad p=1$$

$$L = \frac{1}{\sqrt{\left(\frac{2f}{c}\right)^2 - \left(\frac{1}{a}\right)^2}} = \frac{1}{\sqrt{\left(\frac{2 \cdot 10^{10}}{3 \cdot 10^8}\right)^2 - \left(\frac{1}{0,06}\right)^2}} = \underline{\underline{0,0155 \text{ m}}}$$

TM₁₁₀:

$$f_{\text{REZ}} = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{L}\right)^2} = \frac{3 \cdot 10^8}{2} \sqrt{\left(\frac{1}{0,06}\right)^2 + \left(\frac{1}{0,03}\right)^2} = 5,59 \text{ GHz}$$

Dutínový rezonátor



$$a > b > L$$

$$a = 30 \text{ mm}$$

$$b = 15 \text{ mm}$$

$$L = 10 \text{ mm}$$

a) Nejmenší f_{REZ} ?

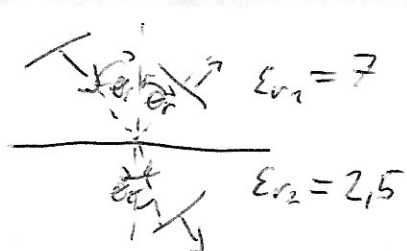
b) Pro mod 20) nakreslete tvar silnicí magn. pole.

$$f_{\text{REZ}} = \frac{c}{2\sqrt{\epsilon_r}} \sqrt{\left(\frac{m}{\underset{\substack{\uparrow \\ 30 \text{ mm}}}{a}}\right)^2 + \left(\frac{n}{\underset{\substack{\uparrow \\ 15 \text{ mm}}}{b}}\right)^2 + \left(\frac{p}{\underset{\substack{\uparrow \\ 10 \text{ mm}}}{L}}\right)^2}$$

To tam snad nebude

REM - limit odrazu?

(0°, 20°, θ_{BR} , θ_{KR} , 50°, 80°)



$$\theta_{BR} = \arcsin\left(\sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1} + \epsilon_{r2}}}\right) = \arcsin\sqrt{\frac{2.5}{9.5}} = 30.86^\circ$$

$$\theta_{KR} = \arcsin\left(\sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1}}}\right) = \arcsin\sqrt{\frac{2.5}{7}} = 0.6405 = 36.7^\circ$$

$$\theta_I \neq \theta_T \quad k_T \sin \theta_T = k_I \sin \theta_I$$

$$\sqrt{\epsilon_2} \sin \theta_T = \sqrt{\epsilon_1} \sin \theta_I$$

$$\sin \theta_T = \frac{\sqrt{\epsilon_1}}{\sqrt{\epsilon_2}} \sin \theta_I =$$

$$\theta_T = \arcsin\left(\frac{\sqrt{\epsilon_1}}{\sqrt{\epsilon_2}} \sin \theta_I\right) =$$

$$\left\{ \begin{array}{ll} \theta_I = 0^\circ & \theta_T = 0^\circ \\ \theta_I = 20^\circ & \theta_T = 34.91^\circ \\ \theta_I = \theta_{BR} & \theta_T = 59.13^\circ \\ \theta_I = \theta_{KR} & \theta_T = 90^\circ \\ \theta_I = 50^\circ & \theta_T = \text{X neprojde nic} \\ \theta_I = 80^\circ & \theta_T = \text{X neprojde nic} \end{array} \right.$$

$$R_{\text{kolmá}}(\theta_I) = \frac{\sqrt{\epsilon_1} \cos \theta_I - \sqrt{\epsilon_2} \cos \theta_T}{\sqrt{\epsilon_1} \cos \theta_I + \sqrt{\epsilon_2} \cos \theta_T}$$

$$R_{\text{rovnooběžná}}(\theta_I) = \frac{\sqrt{\epsilon_1} \cos \theta_T - \sqrt{\epsilon_2} \cos \theta_I}{\sqrt{\epsilon_1} \cos \theta_T + \sqrt{\epsilon_2} \cos \theta_I}$$

$$\theta_I = 0^\circ \quad \theta_T = 0^\circ \quad R_{\text{kolmá}} = -0.2519 \quad R_{\text{rovnooběžná}} = +0.2519$$

$$\theta_I = 20^\circ \quad \theta_T = 34.91^\circ \quad R_{\text{kolmá}} = +0.1871 \quad R_{\text{rovnooběžná}} = 0.3145$$

$$\theta_I = 30.86^\circ = \theta_{BR} \quad \theta_T = 59.13^\circ \quad R_{\perp} = 0.4736 \quad R_{\parallel} = 0$$

$$\theta_I = \theta_{KR} = 36.7^\circ \quad \theta_T = 90^\circ \quad R_{\perp} = +1 \quad R_{\parallel} = -1$$

$$\theta_I = 50^\circ \quad \theta_T = \text{X} \quad |R_{\perp}| = +1 \quad |R_{\parallel}| = +1$$

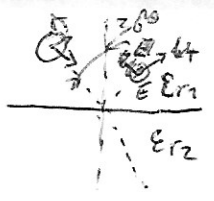
$$\theta_I = 80^\circ \quad \theta_T = \text{X} \quad |R_{\perp}| = +1 \quad |R_{\parallel}| = +1$$

$$\frac{e^{jx} - e^{-jx}}{2j} = y \quad a^{+z} = b^{-z}$$

$$2jy = e^{jx} - e^{-jx}$$

Šikmý dopad

1) Kruhově pol. vlna dopadá silno z diel. $\epsilon_{r1}=4$ pod $\theta_I=36^\circ$
 Odráží se lin pol. vlna



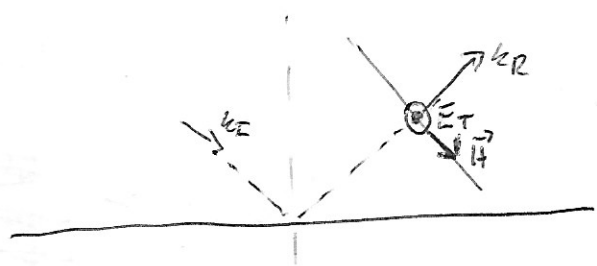
$$\sin \theta_{BR} = \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1} + \epsilon_{r2}}} \quad \Rightarrow \quad \epsilon_{r1} \sin^2 \theta_{BR} = \epsilon_{r2} (1 - \sin^2 \theta_{BR})$$

$$\sin^2 \theta_{BR} = \frac{\epsilon_{r2}}{\epsilon_{r1} + \epsilon_{r2}} \quad \Rightarrow \quad \epsilon_{r1} \frac{\sin^2 \theta_{BR}}{\cos^2 \theta_{BR}} = \epsilon_{r2} = \epsilon_{r1} \tan^2 \theta_{BR}$$

e) $\epsilon_{r2} = 4 \cdot 5,279 \cdot 10^{-1} = \underline{\underline{2,111}}$ ✓

b) Jedná se o kolmou polarizaci

c) $\theta_{KR} = \arcsin \left(\frac{\epsilon_{r2}}{\epsilon_{r1}} \right) = \arcsin \left(\frac{2,111}{4} \right)$
 $= 46,59^\circ$ ✓



② RHEM $F=350 \text{ MHz}$

a) TOT. ODRAZ?

$\theta_{KR} = \arcsin \sqrt{\frac{2}{4}} = 45^\circ$

b) $\lambda_{evanescent}$

a) $\theta_I > \theta_{KR} \Rightarrow$ Dochází k Tot. odrazu

c) $10 \times$ útlum evanescent vlny

b) $\lambda = \frac{2\pi}{k_{zP}} \quad p = \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} \sin \theta_I = 1,225$

c) $0,1 = e^{-10,37z}$

$z = \frac{\ln 0,1}{-10,37} = 0,222 \text{ m}$

$\lambda = \frac{2\pi}{1,225 \cdot 10,37} = 4,946 \cdot 10^{-1} = 0,495 \text{ m}$

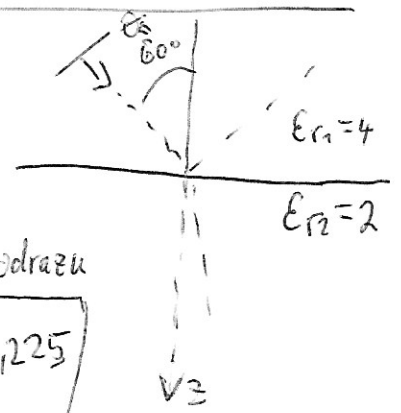
$k_z = \sqrt{-j\omega\mu(j\omega\epsilon + j)} = \omega \sqrt{\mu\epsilon_2} = 2\pi \cdot 350 \cdot 10^6 \sqrt{4\pi \cdot 10^{-7} \cdot 2 \cdot 8,854 \cdot 10^{-12}}$

$= 10,37 \quad k_0 = \frac{2\pi F}{c} = \frac{2\pi \cdot 350 \cdot 10^6}{3 \cdot 10^8} = 7,33 \text{ m}^{-1}$

$k_x = k_1 \sin \theta = k_0 \sqrt{\epsilon_{r1}} \sin \theta = 7,33 \cdot 2 \cdot 0,866 = 12,7 \text{ m}^{-1}$

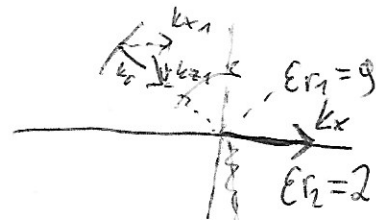
$z = \frac{\ln(10)}{7,33} = 0,314 \text{ m}$

$e^{-k_z z} = \frac{1}{10}$



③ $F = 5 \text{ GHz}$

a) $\theta_{kr} = \arcsin \sqrt{\frac{\epsilon_2}{\epsilon_1}} = \arcsin \sqrt{\frac{2}{9}} = 0,4909$
 $= 28,43^\circ$



b) $0,6 = e^{-k_{z_2} z}$ $z = 2 \text{ mm}$

$k_{z_2} = \frac{\ln 0,6}{-0,002} = 255,4$

$k_x = \sqrt{k_1^2 - k_{z_2}^2}$

$k_x = \sqrt{k_1^2 - k_{z_2}^2}$

$k_x = \sqrt{255,4^2 + 148,7^2} = 295,2$

$k_1 = \omega \sqrt{\epsilon_1 \mu_0} = 2\pi \cdot 5 \cdot 10^9 \cdot \sqrt{9 \cdot 8,854 \cdot 10^{-12}} = 314,4$

$k_x = k_1 \sin \theta_I$

$k_2 = \frac{\omega}{c} \sqrt{\epsilon_2} = \frac{2\pi \cdot 5 \cdot 10^9}{3 \cdot 10^8} \cdot \sqrt{2} = 148,7$

$k_1 = \frac{\omega}{c} \sqrt{\epsilon_1} = \frac{2\pi \cdot 5 \cdot 10^9}{3 \cdot 10^8} \cdot 3 = 314,2$

$\sin \theta_I = \frac{k_x}{k_1} = \frac{295,2}{314,2} = 0,9395$
 $= \frac{295,2}{314,2} = 70^\circ = \theta_I$

$\lambda = \frac{c}{f} = \frac{2\pi}{k_x} = \frac{2\pi}{314,2 \cdot \sin 70^\circ}$
 $= 0,02128 \text{ m}$

④ $\theta_I = 42^\circ$... dopadá ϵ_{r1} na ϵ_{r2} , dochází k polarizaci = Brewster

a) $\frac{\epsilon_{r1}}{\epsilon_{r2}} = ?$

$$\theta_{BR} = \arcsin \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1} + \epsilon_{r2}}}$$

$$\rightarrow \sin^2 \theta_{BR} = \frac{1}{1 + \frac{\epsilon_{r1}}{\epsilon_{r2}}}$$

$$1 + \frac{\epsilon_{r1}}{\epsilon_{r2}} = \frac{1}{\sin^2 \theta_{BR}}$$

$$\frac{\epsilon_{r1}}{\epsilon_{r2}} = \frac{1}{\sin^2 \theta_{BR}} - 1$$

a) = ~~1000~~ $\frac{1}{\sin^2(42^\circ)} - 1 = \underline{\underline{1,233}}$ ✓

b) $E_{im} \dots 100\%$
 $E_{rm} \dots ?\%$

↓
 $80,4\%$

$$R_I = \frac{\sqrt{\epsilon_{r1}} \cos \theta_I - \sqrt{\epsilon_{r2}} \cos \theta_T}{\sqrt{\epsilon_{r1}} \cos \theta_I + \sqrt{\epsilon_{r2}} \cos \theta_T} = \frac{\sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} \cos \theta_I - \cos \theta_T}{\sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} \cos \theta_I + \cos \theta_T} = \frac{\sqrt{1,233} \cos 42^\circ - 0,4183}{\sqrt{1,233} \cos 42^\circ + 0,4183} = \underline{\underline{0,3272}}$$

$$\sqrt{\epsilon_{r1}} \sin \theta_I = \sqrt{\epsilon_{r2}} \sin \theta_T$$

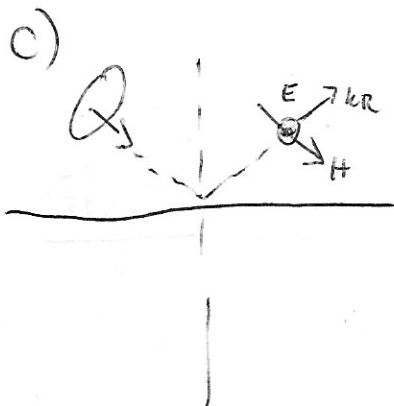
$$\sin \theta_T = \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} \sin \theta_I = \sqrt{1 - \cos^2 \theta_T}$$

$$\cos \theta_T = \sqrt{1 - \frac{\epsilon_{r1}}{\epsilon_{r2}} \sin^2 \theta_I} = 0,4183$$

$$\sin \theta_T = \sqrt{1,233} \sin 42^\circ = 0,743$$

$$\theta_T = 0,8375$$

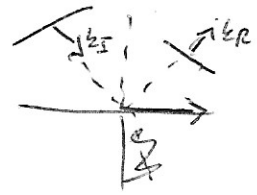
$$\cos \theta_T = 0,6693$$



5. příklad vhy odraz

$$\epsilon_{r1} = 5 \quad \epsilon_{r2} = 1,5 \quad f = 15 \text{ GHz}$$

Totální odraz



$$a) \theta_{kr} = ? = \arcsin \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1}}} = \arcsin \sqrt{\frac{1,5}{5}} = 0,5496 = 17,469^\circ = 33,21^\circ \checkmark$$

$$b) \lambda = 0,01 \text{ m}$$

$$k_{r1} = \frac{\omega}{c} \sqrt{\epsilon_{r1}} = \frac{2\pi \cdot 15 \cdot 10^9}{3 \cdot 10^8} \sqrt{5} = 702,5$$

$$\lambda = \frac{2\pi}{k_x} = \frac{2\pi}{k_{r1} \sin \theta_I}$$

$$\sin \theta_I = \frac{2\pi}{k_{r1} \lambda} = \frac{2\pi}{702,5 \cdot 0,01} = 0,8944$$

$$\theta_I = 63,43^\circ \checkmark$$

$$k_x = k_{r1} \sin \theta_I = 702,5 \cdot 0,8944 = 628,3 \text{ m}^{-1}$$

$$c) z = ? \text{ útlumi } 15\%$$

$$e^{-\gamma z} = 0,15$$

$$z = \frac{\ln 0,15}{-\gamma}$$

$$= \frac{\ln 0,15}{-496,7} = 3,819 \text{ mm} \checkmark$$

$$\gamma = \sqrt{k_x^2 - k_2^2}$$

$$\gamma = \sqrt{628,3^2 - 384,8^2} = 496,7 \text{ m}^{-1}$$

$$k_2 = \frac{\omega}{c} \sqrt{\epsilon_{r2}} =$$

$$= \frac{2\pi \cdot 15 \cdot 10^9 \cdot \sqrt{1,5}}{3 \cdot 10^8}$$

$$= 384,8 \text{ m}^{-1}$$